

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 9

Date: Oct 19, 2023

1. For the emitter-bias network in the active region (base-emitter forward bias, base-collector reverse bias) Fig.1, let $V_{CC} = +18V$, $\beta = 60$, $R_B = 430k\Omega$, $R_C = 2k\Omega$, $R_E = 1k\Omega$. Determine the following (assume $V_{BE} = 0.7V$):

- (a) I_B
- (b) I_C
- (c) V_{CE}
- (d) V_C
- (e) V_E
- (f) V_B
- (g) V_{BC}

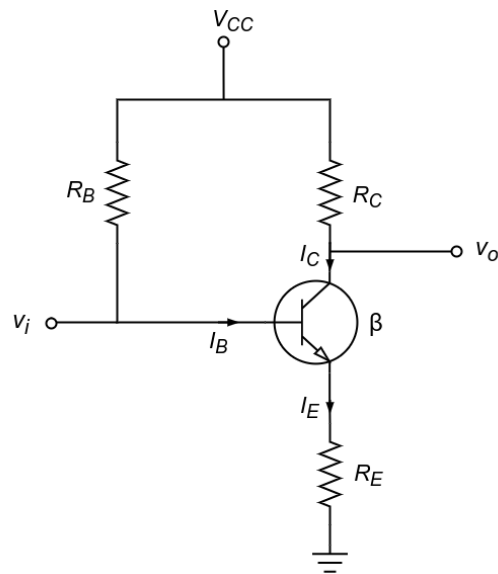


Figure 1

2. Determine the region where the transistor is operating. Accordingly, compute the base current and the collector current. Let $V_{CC} = 10V$, $V_{BB} = 5V$, $R_C = 12k\Omega$, $R_B = 50k\Omega$, $R_E = 5k\Omega$. Assuming $\beta = 60$, $V_{BE} = 0.7V$ and V_{CE} at saturation is $0.1V$

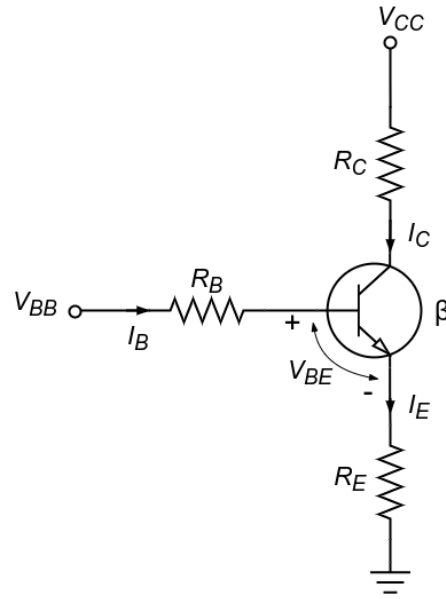


Figure 2

3. Calculate the line currents in the three-wire $Y - Y$ system of Figure 3.

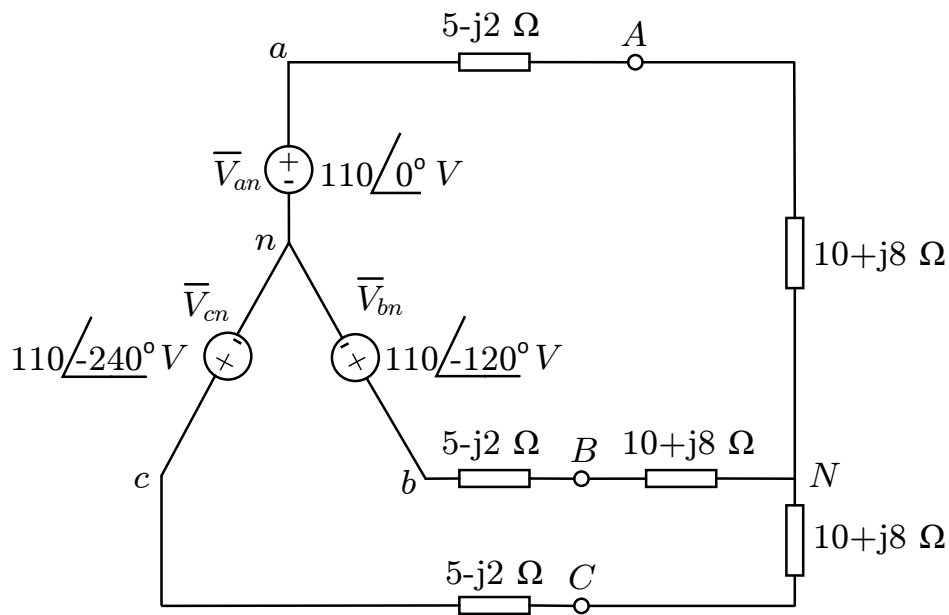


Figure 3 Three-wire $Y - Y$ system.

4. A balanced abc -sequence Y -connected source with $\bar{V}_{an} = 100/\underline{10^\circ}$ V is connected to a Δ -connected balanced load $(8 + j4) \Omega$ per phase as shown in Figure 4. Calculate the phase and line currents.

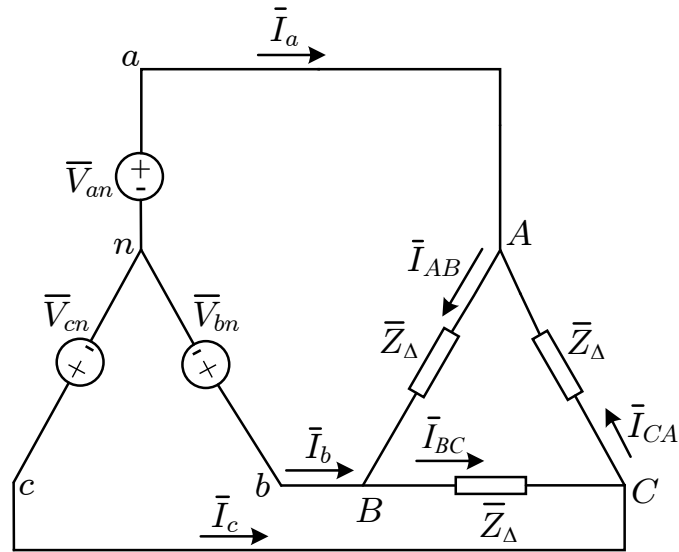


Figure 4 Three-wire $Y - \Delta$ system.

5. A balanced three-phase system with a line voltage of 300 V supplies a balanced Y-connected load with 1200 W at a leading PF of 0.8. Find the line current and the per-phase load impedance. Further, a balanced 600 W lighting load is added (in parallel) to the system below. Determine the new line current.